

GPIO Inputs – Control Buttons (polled, not interrupt)

Pin	Signal	Mode	Pull
PA0	Toggle Direction	GPIO_Input	Pull-up
PA1	+RPM	GPIO_Input	Pull-up
PA2	-RPM	GPIO_Input	Pull-up
PA3	+Microstep	GPIO_Input	Pull-up
PA4	-Microstep	GPIO_Input	Pull-up

GPIO Outputs – A4988 Driver Control

Pin	Signal	Mode	Output Type	Pull	Speed
PA5	STEP	GPIO_Output	Push-Pull	No pull	Medium
PA6	DIRECTION	GPIO_Output	Push-Pull	No pull	Low
PA7	MS1	GPIO_Output	Push-Pull	No pull	Low
PB0	MS2	GPIO_Output	Push-Pull	No pull	Low
PB1	MS3	GPIO_Output	Push-Pull	No pull	Low

GPIO Input – E-Stop (interrupt-driven)

Setting	Value
Pin	PB2
Mode	GPIO_EXTI2
Trigger	Falling edge
Pull	Pull-up (internal)
NVIC	Note: Enable EXTI2_3_IRQn, priority numerically >= configLIBRARY_MAX_SYSCALL_INTERRUPT_PRIORITY

ST7735 Display

Pin	Signal	Mode	Notes
PB3	SCK	SPI1 alternate function	Auto-configured once SPI1 enabled
PA12	MOSI	SPI1 alternate function	Confirm CubeMX shows this as a valid AF option for SPI1_MOSI on the package – auto-assigned once SPI1 mode is set
PB7	A0 (DC)	GPIO_Output	Push-Pull, No pull
PB8	RESET	GPIO_Output	Push-Pull, No pull
PB9	CS	GPIO_Output	Push-Pull, No pull

SPI1 peripheral settings:

Setting	Value
Mode	Transmit Only Master
Clock Polarity (CPOL)	Low (Mode 0 - verify against your specific ST7735 breakout if display doesn't work)
Clock Phase (CPHA)	1st Edge
Data Size	8 bits
First Bit	MSB First

TIM3 (microsecond delay counter)

Setting	Value
Clock Source	Internal Clock
Channels	All Disabled
Prescaler	11
Counter Mode	Up
Auto-Reload	0xFFFF
Auto-reload preload	Disable
NVIC	Disabled

TIM16 (step-pulse generator)

Setting	Value
Clock Source	Internal Clock
Channels	Disabled
Prescaler	11
Counter Mode	Up
Auto-Reload	Placeholder (e.g. 1000) - overwritten at runtime
Auto-reload preload	Enable
NVIC	Enabled